

ALESSANDRO TRIFILETTI

professor

基本信息

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学科领域

集成电路设计与集成系统

微电子科学与工程

研究领域

低功耗电路与能量管理

模拟与射频IC设计

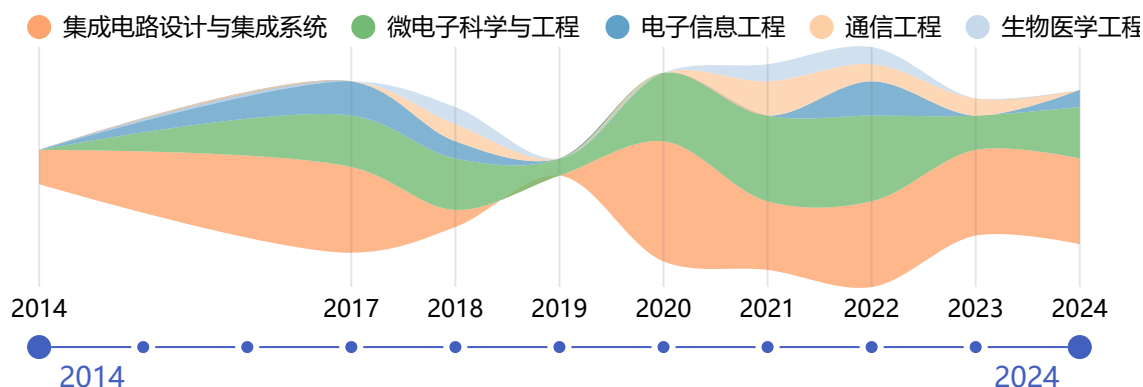
混合信号集成技术

模拟与数模混合电路

个人简介

Alessandro Trifiletti, born on October 4, 1959, in Rome, is a distinguished professor at the Faculty of Engineering at the University of Rome "La Sapienza." He began his career as a qualified technician in 1991, focusing on research in analog, RF, and microwave IC projects. Trifiletti has made significant contributions to microelectronics, including statistical modeling of III-V active devices, circuit theory, robust integrated circuit design methodologies, and multifunctional, RF, and mixed-signal IC development. He has co-authored approximately 280 publications, with over 130 in international journals, achieving an h-index of 24. Trifiletti currently serves as a reviewer for IEEE and IEE journals.

近年发表的论文



A novel FVF-based GHz-range biquad in a 28 nm CMOS FD-SOI technology

AEU-INTERNATIONAL JOURNAL OF ELECTRONICS AND COMMUNICATIONS

Lombardo, Matteo; Centurelli, Francesco; Monsurro, Pietro; Trifiletti, Alessandro

被引用次数: 0

2024

A 150 MS/s, 10 bit SAR ADC Featuring a Modified Quasi-Monotonic Switching Scheme

2024 19TH CONFERENCE ON PH.D RESEARCH IN MICROELECTRONICS AND ELECTRONICS, PRIME 2024

Spinogatti, Valerio; Bocciarelli, Cristian; Eusebio, Lorenzo; Centurelli, Francesco; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 0

2024

Enhancing Performance of Ultra-Low Voltage Body-Driven Comparators through Clocked Supply Voltage

2024 19TH CONFERENCE ON PH.D RESEARCH IN MICROELECTRONICS AND ELECTRONICS, PRIME 2024

Della Sala, Riccardo; Bocciarelli, Cristian; Spinogatti, Valerio; Centurelli, Francesco; Trifiletti, Alessandro

被引用次数: 0

2024

A 0.3 V OTA with Enhanced CMRR and High Robustness to PVT Variations

JOURNAL OF LOW POWER ELECTRONICS AND APPLICATIONS

Della Sala, Riccardo; Centurelli, Francesco; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 0

2024

An Intermodulation Radar for Non-Linear Target and Transceiver Detection

SENSORS

Pisa, Stefano; Trifiletti, Alessandro; Tommasino, Pasquale; Monsurro, Pietro; Tognolatti, Piero; Leuzzi, Giorgio; Di Carlofelice, Alessandro; Di Giampaolo, Emidio

被引用次数: 0

2024

Body Biasing Techniques for Dynamic Comparators: A Systematic Survey

ELECTRONICS

Spinogatti, Valerio; Della Sala, Riccardo; Bocciarelli, Cristian; Centurelli, Francesco; Trifiletti, Alessandro

被引用次数: 0

2024

Exploiting Body-Driven Feedbacks in Physical Unclonable Functions for Ultra Low Voltage, Ultra Low Power Applications: A 0.3 V Weak-PUF

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Della Sala, Riccardo; Bellizia, Davide; Centurelli, Francesco; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 0

2024

A 0.15-to-0.5 V Body-Driven Dynamic Comparator with Rail-to-Rail ICMR

JOURNAL OF LOW POWER ELECTRONICS AND APPLICATIONS

Della Sala, Riccardo; Spinogatti, Valerio; Bocciarelli, Cristian; Centurelli, Francesco; Trifiletti, Alessandro

被引用次数: 8

2023

A body-driven rail-to-rail 0.3 V operational transconductance amplifier exploiting current gain stages

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Della Sala, Riccardo; Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 1

2023

An Improved Strong Arm Comparator With Integrated Static Preamplifier

IEEE ACCESS

Spinogatti, Valerio; Sala, Riccardo Della; Bocciarelli, Cristian; Centurelli, Francesco; Trifiletti, Alessandro

被引用次数: 4

2023

A Novel Parallel Digitizer With a Pulseless Mixing-Filtering-Processing Architecture and Its Implementation in a SiGe HBT Technology at 40GS/s

IEEE ACCESS

Centurelli, Francesco; Monsurro, Pietro; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 2

2023

A 0.3V Rail-to-Rail Three-Stage OTA With High DC Gain and Improved Robustness to PVT Variations

IEEE ACCESS

Della Sala, Riccardo; Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 14

2023

High-Accuracy Low-Cost Generalized Complex Pruned Volterra Models for Nonlinear Calibration

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Bocciarelli, Cristian; Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Spinogatti, Valerio; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 1

2023

General Approach to the Calibration of Innovative MFP Multichannel Digitizers

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Centurelli, Francesco; Monsurro, Pietro; Trifiletti, Alessandro; D'Arco, Mauro; Angrisani, Leopoldo

被引用次数: 5

2022

A Tree-Based Architecture for High-Performance Ultra-Low-Voltage Amplifiers

JOURNAL OF LOW POWER ELECTRONICS AND APPLICATIONS

Centurelli, Francesco; Della Sala, Riccardo; Monsurro, Pietro; Scotti, Giuseppe; Trifiletti, Alessandro

A Differential-to-Single-Ended Converter Based on Enhanced Body-Driven Current Mirrors Targeting Ultra-Low-Voltage OTAs

ELECTRONICS

Sala, Riccardo Della; Centurelli, Francesco; Scotti, Giuseppe; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 11

2022

Methods for Model Complexity Reduction for the Nonlinear Calibration of Amplifiers Using Volterra Kernels

ELECTRONICS

Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 1

2022

A New Fully Closed-Loop, High-Precision, Class-AB CCII for Differential Capacitive Sensor Interfaces

ELECTRONICS

Barile, Gianluca; Centurelli, Francesco; Ferri, Giuseppe; Monsurro, Pietro; Pantoli, Leonardo; Stornelli, Vincenzo; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 1

2022

80 dB tuning range transimpedance amplifier exploiting the Switched-Resistor approach

AEU-INTERNATIONAL JOURNAL OF ELECTRONICS AND COMMUNICATIONS

Centurelli, Francesco; Fava, Alessandro; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 4

2022

An Ultra-Low-Voltage class-AB OTA exploiting local CMFB and Body-to-Gate interface

AEU-INTERNATIONAL JOURNAL OF ELECTRONICS AND COMMUNICATIONS

Centurelli, Francesco; Della Sala, Riccardo; Monsurro, Pietro; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 19

2022

A SiGe HBT 6th-Order 10 GHz Inductor-Less Anti-Aliasing Low-Pass Filter for High-Speed ATI Digitizers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 7

2022

Low power class-AB VCII with extended dynamic range

AEU-INTERNATIONAL JOURNAL OF ELECTRONICS AND COMMUNICATIONS

Barile, Gianluca; Ferri, Giuseppe; Pantoli, Leonardo; Ragnoli, Mattia; Stornelli, Vincenzo; Safari, Leila; Centurelli, Francesco; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 6

2022

A 0.3 V, Rail-to-Rail, Ultralow-Power, Non-Tailed, Body-Driven, Sub-Threshold Amplifier

APPLIED SCIENCES-BASEL

Centurelli, Francesco; Della Sala, Riccardo; Scotti, Giuseppe; Trifiletti, Alessandro

An E-band Variable Gain Amplifier with 24 dB-control range and 80 to 100 GHz 1 dB bandwidth in SiGe BiCMOS technology

FREQUENZ

Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 1

2021

A Novel OTA Architecture Exploiting Current Gain Stages to Boost Bandwidth and Slew-Rate

ELECTRONICS

Centurelli, Francesco; Della Sala, Riccardo; Monsurro, Pietro; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 13

2021

Distributed switched-resistor approach for high-Q biquad filters

AEU-INTERNATIONAL JOURNAL OF ELECTRONICS AND COMMUNICATIONS

Centurelli, Francesco; Fava, Alessandro; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 5

2021

Design of Low-Voltage Power Efficient Frequency Dividers in Folded MOS Current Mode Logic

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Centurelli, Francesco; Scotti, Giuseppe; Trifiletti, Alessandro; Palumbo, Gaetano

被引用次数: 5

2021

A class-AB linear transconductor with enhanced linearity

AEU-INTERNATIONAL JOURNAL OF ELECTRONICS AND COMMUNICATIONS

Bocciarelli, Cristian; Centurelli, Francesco; Monsurro, Pietro; Spinogatti, Valerio; Trifiletti, Alessandro

被引用次数: 4

2021

Compact E-Band I/Q Receiver in SiGe BiCMOS for 5G Backhauling Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Amendola, Giandomenico; Boccia, Luigi; Centurelli, Francesco; Chevalier, Pascal; Fonte, Alessandro; Mustacchio, Carmine; Pallotta, Andrea; Tommasino, Pasquale; Traversa, Antonio; Trifiletti, Alessandro

被引用次数: 8

2021

A Power Efficient Frequency Divider With 55 GHz Self-Oscillating Frequency in SiGe BiCMOS

ELECTRONICS

Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 3

2020

A low-voltage class-AB OTA exploiting adaptive biasing

AEU-INTERNATIONAL JOURNAL OF ELECTRONICS AND COMMUNICATIONS

Centurelli, Francesco; Fava, Alessandro; Olivieri, Mauro; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 7

2020

A Novel 0.5 V MCML D-Flip-Flop Topology Exploiting Forward Body Bias Threshold Lowering

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Scotti, Giuseppe; Trifiletti, Alessandro; Palumbo, Gaetano

被引用次数: 17

2020

Delay models and design guidelines for MCML gates with resistor or PMOS load

MICROELECTRONICS JOURNAL

Centurelli, Francesco; Scotti, Giuseppe; Trifiletti, Alessandro; Palumbo, Gaetano

被引用次数: 7

2020

0.6-V CMOS cascode OTA with complementary gate-driven gain-boosting and forward body bias

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Cellucci, Danilo; Centurelli, Francesco; Di Stefano, Valerio; Monsurro, Pietro; Pennisi, Salvatore; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 19

2020

An improved reversed miller compensation technique for three-stage CMOS OTAs with double pole-zero cancellation and almost single-pole frequency response

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 3

2020

Low-power class-AB 4th-order low-pass filter based on current conveyors with dynamic mismatch compensation of biasing errors

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Centurelli, Francesco; Monsurro, Pietro; Stornelli, Vincenzo; Barile, Gianluca; Trifiletti, Alessandro

被引用次数: 1

2020

Two novel architectures for 4-channel mixing/filtering/processing digitizers

MEASUREMENT

Monsurro, Pietro; Trifiletti, Alessandro; Angrisani, Leopoldo; D'Arco, Mauro

被引用次数: 13

2019

New Models for the Calibration of Four-Channel Time-Interleaved ADCs Using Filter Banks

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Monsurro, Pietro; Rosato, Felice; Trifiletti, Alessandro

被引用次数: 22

2018

A Topology of Fully Differential Class-AB Symmetrical OTA With Improved CMRR

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Centurelli, Francesco; Monsurro, Pietro; Parisi, Gaetano; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 18

2018

A Directly Modulated Multiwavelength Transmitter Monolithically Integrated on InP

IEEE JOURNAL OF SELECTED TOPICS IN QUANTUM ELECTRONICS

Andriolli, Nicola; Velha, Philippe; Chiesa, Marco; Trifiletti, Alessandro; Contestabile, Giampiero

被引用次数: 14

2018

Low power DDA-based instrumentation amplifier for neural recording applications in 65 nm CMOS

AEU-INTERNATIONAL JOURNAL OF ELECTRONICS AND COMMUNICATIONS

Avoli, Matteo; Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 26

2018

TEL Logic Style as a Countermeasure Against Side-Channel Attacks: Secure Cells Library in 65nm CMOS and Experimental Results

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Bellizia, Davide; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 13

2018

Streamline calibration modelling for a comprehensive design of ATI-based digitizers

MEASUREMENT

Monsurro, Pietro; Trifiletti, Alessandro; Angrisani, Leopoldo; D'Arco, Mauro

被引用次数: 19

2018

Univariate Power Analysis Attacks Exploiting Static Dissipation of Nanometer CMOS VLSI Circuits for Cryptographic Applications

IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING

Bellizia, Davide; Bongiovanni, Simone; Monsurro, Pietro; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 15

2017

A Novel Framework to Estimate the Path Delay Variability On the Back of an Envelope via the Fan-Out-of-4 Metric

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Alioto, Massimo; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 22

2017

The AB-CCII, a novel adaptive biasing LV-LP current conveyor architecture

AEU-INTERNATIONAL JOURNAL OF ELECTRONICS AND COMMUNICATIONS

Stornelli, V.; Pantoli, L.; Ferri, G.; Liberati, L.; Centurelli, F.; Monsurro, P.; Trifiletti, A.

被引用次数: 9

2017

Design of Low-Voltage High-Speed CML D-Latches in Nanometer CMOS Technologies

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Scotti, Giuseppe; Bellizia, Davide; Trifiletti, Alessandro; Palumbo, Gaetano

被引用次数: 24

2017

Faster, Stabler, and Simpler-A Recursive-Least-Squares Algorithm Exploiting the Frisch-Waugh-Lovell Theorem

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Monsurro, Pietro; Trifiletti, Alessandro

被引用次数: 7

2017

Calibration of Time-Interleaved ADCs via Hermitianity-Preserving Taylor Approximations

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Monsurro, Pietro; Trifiletti, Alessandro

被引用次数: 15

2017

0.9-V Class-AB Miller OTA in 0.35- μ m CMOS With Threshold-Lowered Non-Tailed Differential Pair

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Grasso, Alfio Dario; Pennisi, Salvatore; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 48

2017

Effectiveness of Leakage Power Analysis Attacks on DPA-Resistant Logic Styles Under Process Variations

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Alioto, Massimo; Bongiovanni, Simone; Djukanovic, Milena; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 55

2014

88- μ A 1-MHz Stray-Insensitive CMOS Current-Mode Interface IC for Differential Capacitive Sensors

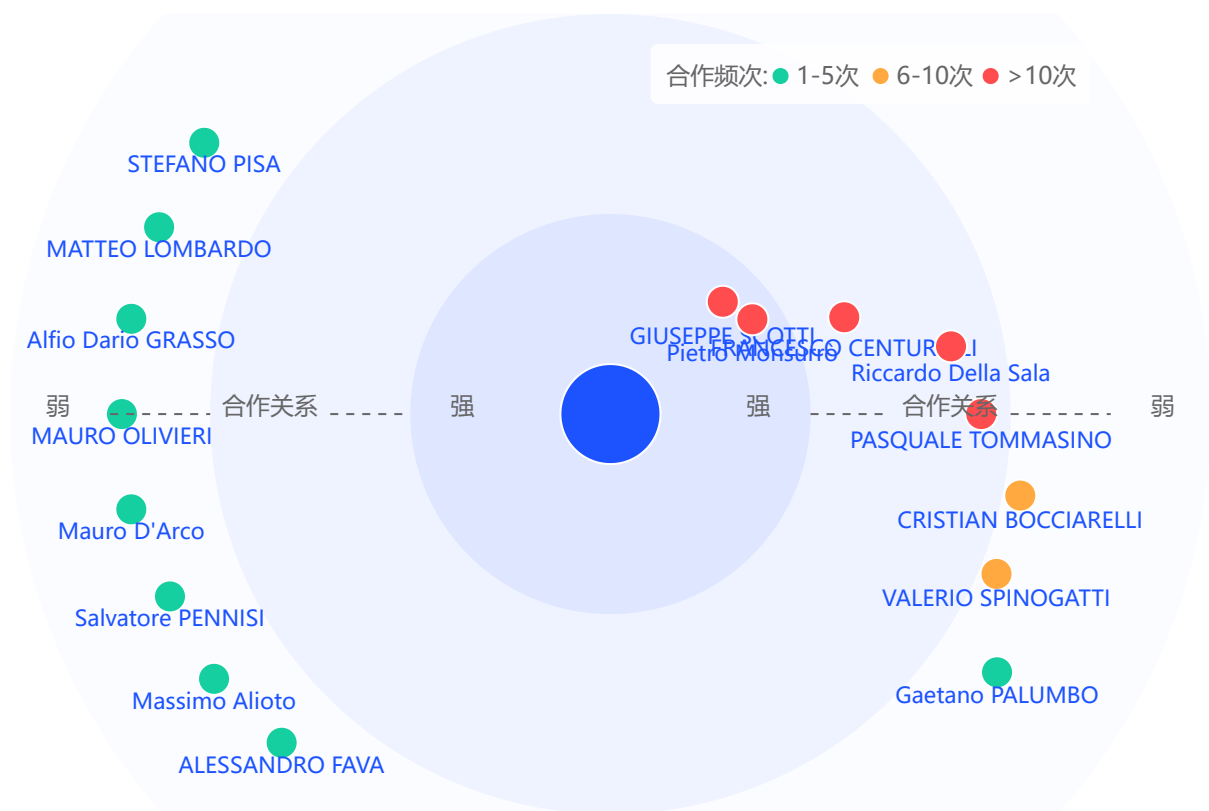
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Scotti, Giuseppe; Pennisi, Salvatore; Monsurro, Pietro; Trifiletti, Alessandro

被引用次数: 41

2014

学术合作



GIUSEPPE SCOTTI

罗马第一大学

合作次数: 29

Pietro Monsurrò

罗马第一大学

合作次数: 28

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